

Go-Around Detection at KDAB — Summary Report

ERAU fleet arrivals, 2025-01-02 to 2025-12-23 · ADS-B based detection pipeline (goaround_pipeline)

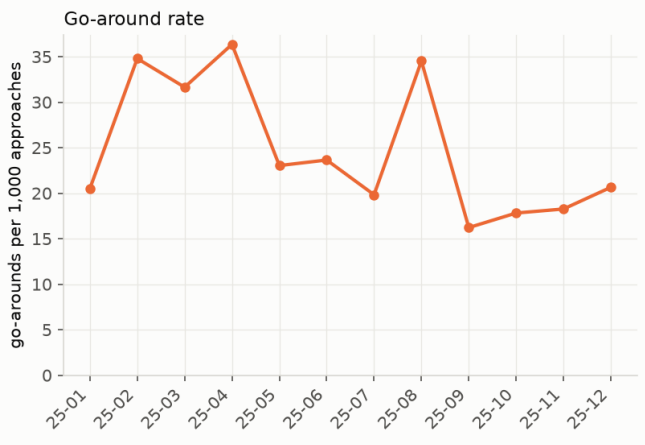
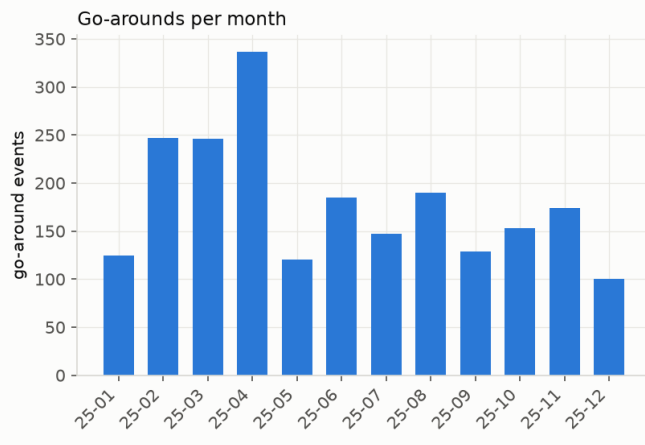
1,888 go-around events were identified among **87,092 independent approach attempts** (91,810 aligned segments including continued fragments) — a rate of **21.7 per 1,000 approaches** under the strict definition. A further 265 candidates with borderline low-point geometry are retained and flagged as ambiguous.

Outcome breakdown

Outcome	Count	Share
Full-stop landing	60,090	65.5%
Touch-and-go	19,095	20.8%
Continued approach (fragment)	4,718	5.1%
Unresolved	4,449	4.8%
Go-around	1,888	2.1%
Low approach	1,305	1.4%
Ambiguous go-around candidate	265	0.3%

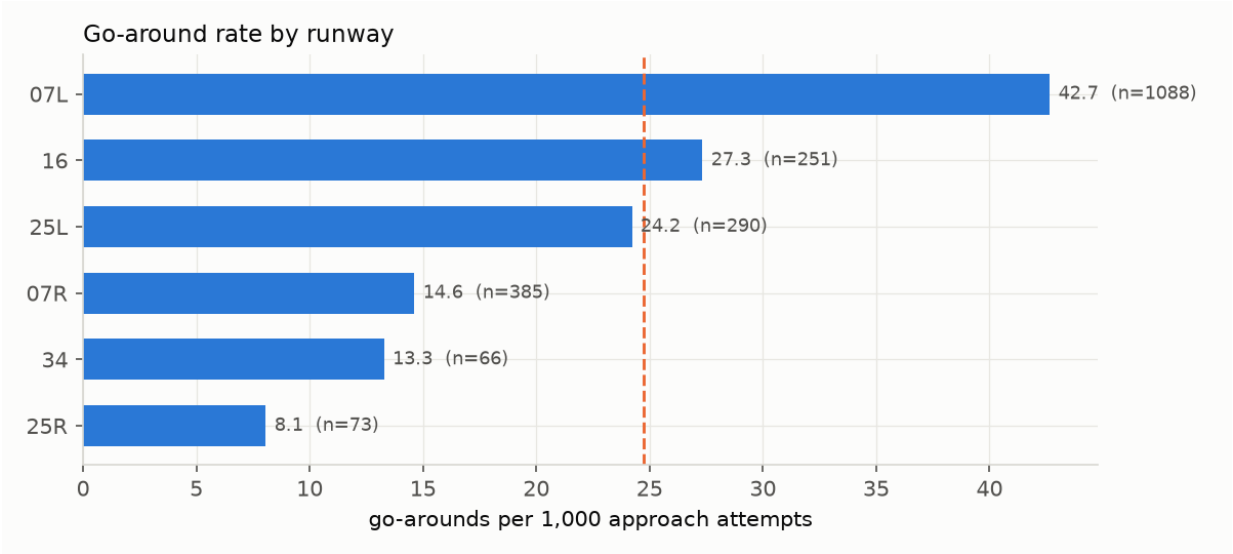
Monthly

Month	Approaches	Go-arounds	Rate /1,000
2025-01	6,092	125	20.5
2025-02	7,099	247	34.8
2025-03	7,774	246	31.6
2025-04	9,271	337	36.3
2025-05	5,209	120	23.0
2025-06	7,824	185	23.6
2025-07	7,413	147	19.8
2025-08	5,504	190	34.5
2025-09	7,952	129	16.2
2025-10	8,585	153	17.8
2025-11	9,526	174	18.3
2025-12	4,843	100	20.6



By runway

Runway	Approaches	Go-arounds	Rate /1,000
07R	26,324	385	14.6
07L	25,503	1088	42.7
25L	11,974	290	24.2
16	9,182	251	27.3
25R	9,050	73	8.1
34	4,968	66	13.3



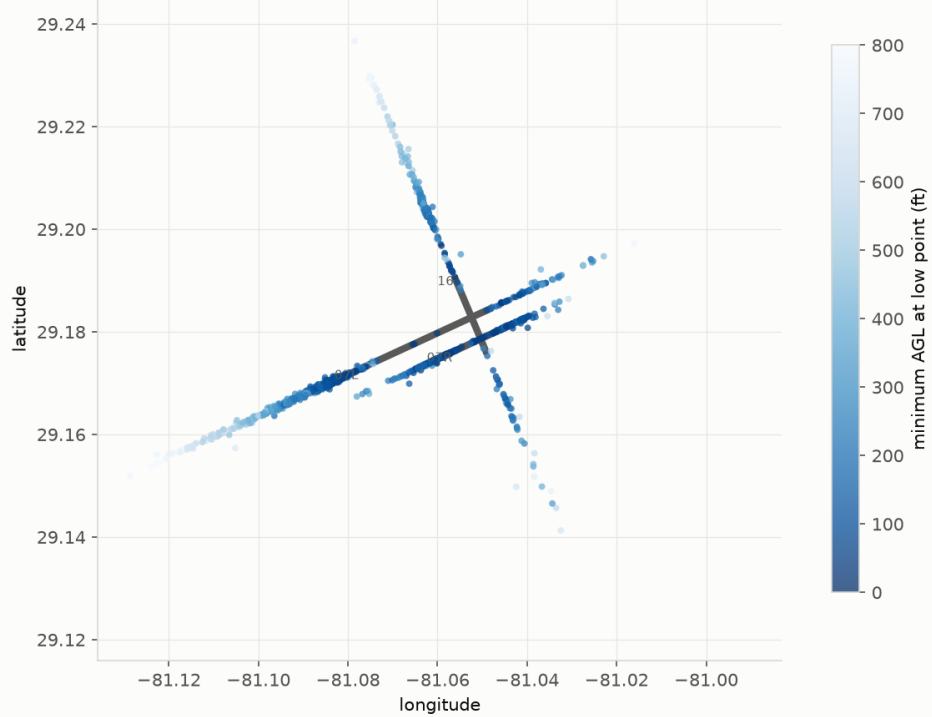
Go-around event characteristics

	Min AGL (ft)	Initiation AGL (ft)	Peak climb (fpm)	Altitude regained (ft)
mean	179	204	1,114	1,485
median	116	141	1,088	1,700
25th pct	91	116	960	875
75th pct	241	266	1,280	1,875
min	41	16	320	250
max	791	941	2,176	3,000

After going around, 1,815 aircraft (96%) flew another approach in the same flight; the rest end their data record (departure or coverage loss).

Where go-arounds begin

Go-around initiation points at KDAB (n=1888)



Method in brief

Detection. Each aircraft-day is split into flight legs on 20-minute gaps. An approach attempt is a ≥ 20 s interval aligned with a runway extended centerline (< 5 NM out, < 0.30 NM cross-track, track within 25° of the runway heading, below 1,500 ft AGL, descending). Parallel-runway assignment uses a low-altitude-weighted vote.

Classification. Touchdown evidence (on-ground flag, $\text{AGL} \leq 25$ ft, a ≥ 20 s flat stretch below 120 ft AGL, groundspeed ≤ 50 kt, or a data dropout below 150 ft) separates landings and touch-and-goes. A go-around is a no-touchdown approach whose low point (below 1,000 ft AGL, after ≥ 150 ft of observed descent) reverses into a sustained climb (≥ 300 fpm for ≥ 15 s, regaining ≥ 250 ft) with ≤ 20 s spent at the profile low point. Level segments of ≥ 30 s classify as deliberate low approaches; 20-30 s cases are flagged ambiguous. The two populations are separated by an empty gap in the plateau-duration distribution (figure below).

Validation. Stratified manual review of event plots; cross-agreement with the traffic library's published go-around detector (all disagreements definitional); one-at-a-time threshold sensitivity: counts stable within $\pm 5\%$ for most parameters, ± 10 -20% only for the two parameters defining the class boundary at flare height.

Limitations. Pilot intent is invisible in surveillance data (practice and directed go-arounds are counted alike, as specified). Go-arounds initiated entirely below the local ADS-B coverage floor (~ 50 -250 ft AGL) are undetectable. Scope is go-arounds from established, runway-aligned finals; circling breakoffs are out of scope.

